

Name: _____

A Level Mathematics : Pure Mathematics

Topic 3 :Coordinate Geometry Circles

Date:

Time:

Total marks available:

Total marks achieved: _____

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Questions

Q1.

The circle C has equation

$$x^2 + y^2 - 10x + 4y + 11 = 0$$

(a) Find

- (i) the coordinates of the centre of C ,
- (ii) the exact radius of C , giving your answer as a simplified surd.

(4)

The line l has equation $y = 3x + k$ where k is a constant.

Given that l is a tangent to C ,

- (b) find the possible values of k , giving your answers as simplified surds.

(5)

(Total for question = 9 marks)

Q2.

A circle has equation

$$x^2 + y^2 - 10x + 16y = 80$$

(a) Find

- (i) the coordinates of the centre of the circle,
- (ii) the radius of the circle.

(3)

Given that P is the point on the circle that is furthest away from the origin O ,

- (b) find the exact length OP

(2)

(Total for question = 5 marks)

Q3.

A circle C has equation

$$x^2 + y^2 - 4x + 10y = k$$

where k is a constant.

(a) Find the coordinates of the centre of C .

(2)

(b) State the range of possible values for k .

(2)

(Total for question = 4 marks)

Q4.

A circle C with radius r

- lies only in the 1st quadrant
- touches the x -axis and touches the y -axis

The line l has equation $2x + y = 12$

(a) Show that the x coordinates of the points of intersection of l with C satisfy

$$5x^2 + (2r - 48)x + (r^2 - 24r + 144) = 0$$

(3)

Given also that l is a tangent to C ,

(b) find the two possible values of r , giving your answers as fully simplified surds.

(4)

(Total for question = 7 marks)

Q5.

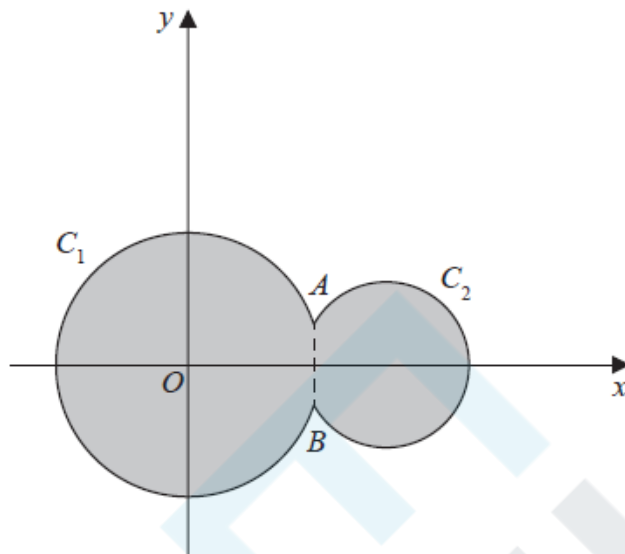


Figure 3

Circle C_1 has equation $x^2 + y^2 = 100$

Circle C_2 has equation $(x - 15)^2 + y^2 = 40$

The circles meet at points A and B as shown in Figure 3.

(a) Show that angle $AOB = 0.635$ radians to 3 significant figures, where O is the origin.

(4)

The region shown shaded in Figure 3 is bounded by C_1 and C_2

(b) Find the perimeter of the shaded region, giving your answer to one decimal place.

(4)

(Total for question = 8 marks)

Q6.

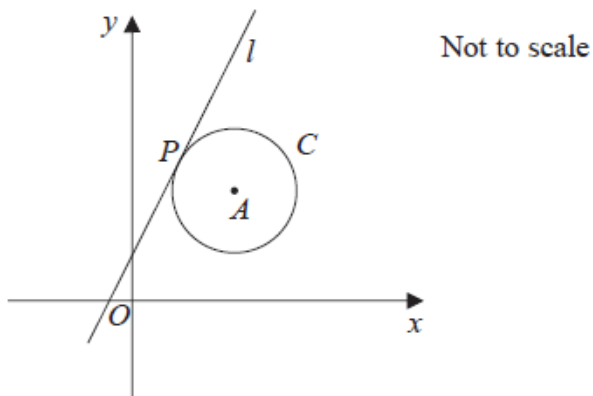


Figure 3

The circle C has centre A with coordinates $(7, 5)$.

The line l , with equation $y = 2x + 1$, is the tangent to C at the point P , as shown in Figure 3.

(a) Show that an equation of the line PA is $2y + x = 17$

(3)

(b) Find an equation for C .

(4)

The line with equation $y = 2x + k$, $k \neq 1$ is also a tangent to C .

(c) Find the value of the constant k .

(3)

(Total for question = 10 marks)